

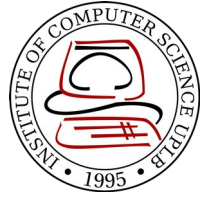
CMSC 137

Data Communications and Networking

Lecture Module
02 – The Internet



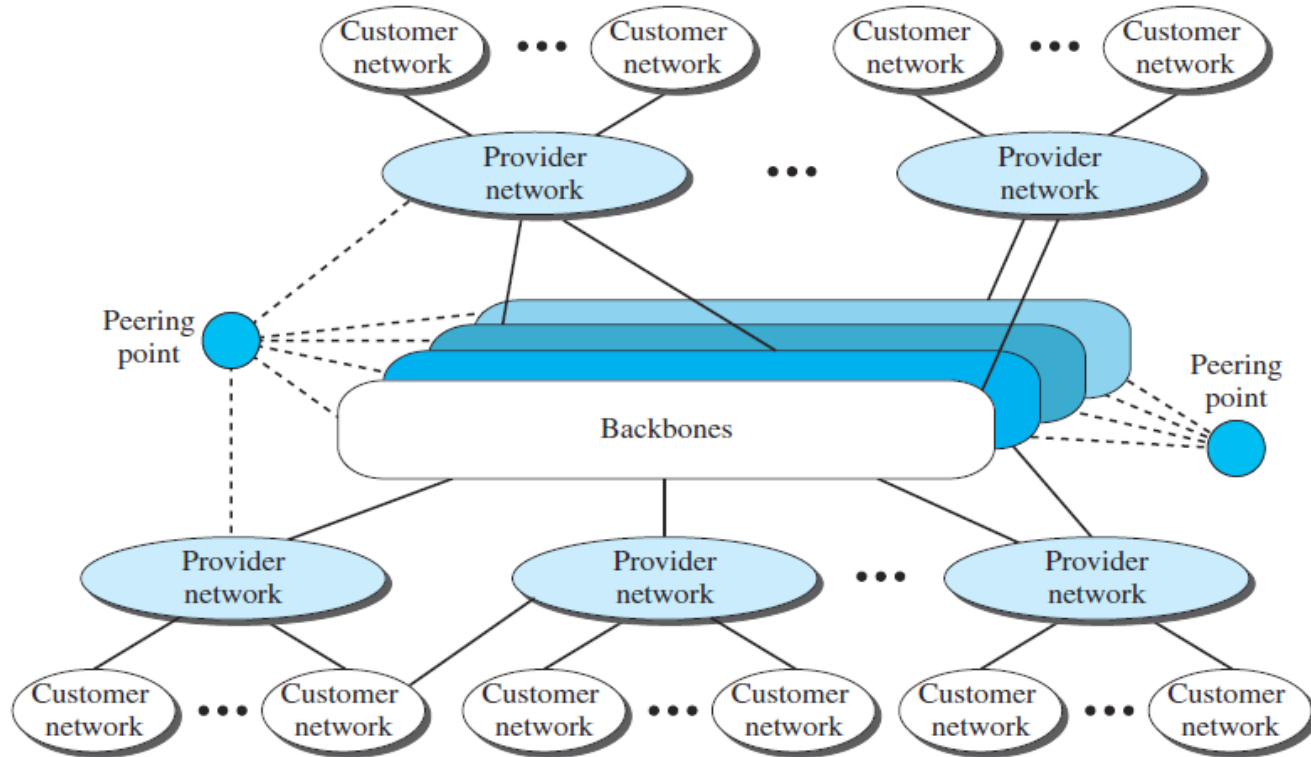
University of the Philippines
LOS BAÑOS



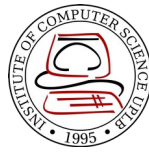
Module 2: The Internet

Rozano S. Maniaol
Lecturer

The Internet

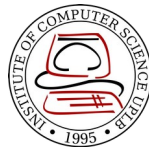


The Internet



- **Backbones** – large networks owned by communication companies
- **Peering points** – complex switching systems
- **Provider networks** – uses the service of the backbone for a fee
- **Customer networks** (at the edge) – actually uses the services provided by the Internet
- Backbones and provider networks are also called **Internet Service Providers** (ISPs)
 - Backbones (international ISP); Provider networks (national or regional ISPs)

Accessing the Internet



- Using telephone networks
 - Dial-up service
 - DSL service
- Using cable networks
- Using wireless networks
- Direct connection to the Internet

History



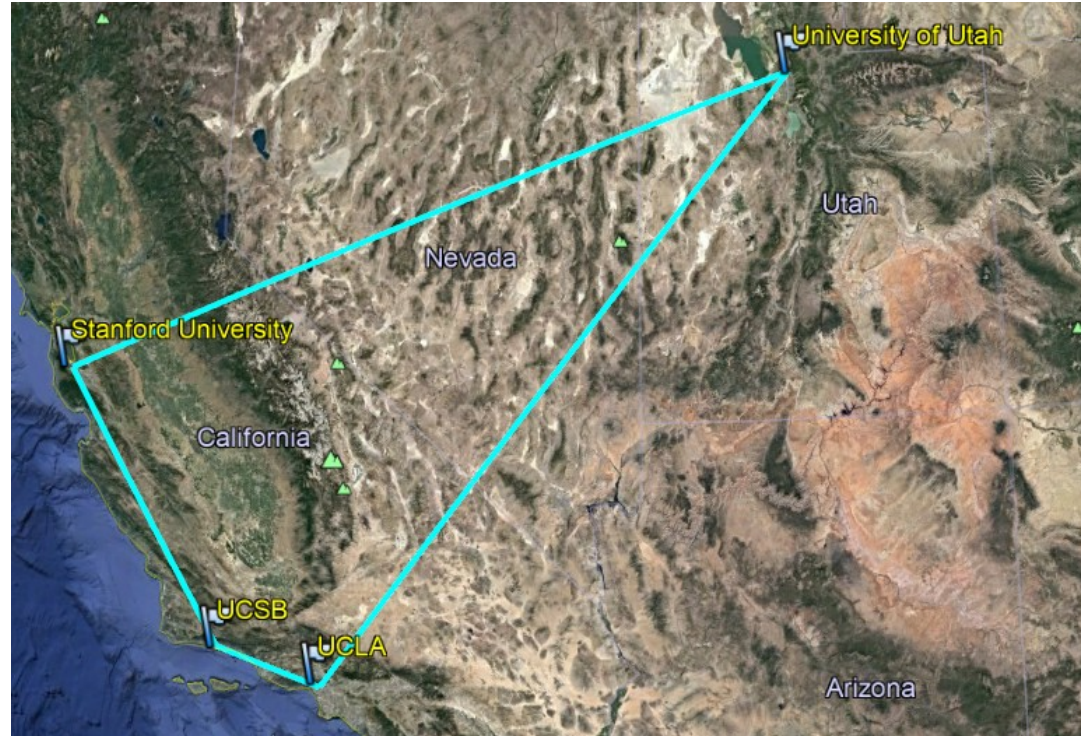
- Mid-1960's – computers are standalone
- **Advanced Research Projects Agency (ARPA)** was interested in finding a way to connect computers to reduce cost and eliminate duplication of efforts
- 1967 – at an ACM meeting, ARPA presented **ARPANET**
 - small network of computers which used **Interface Message Processors (IMP)**



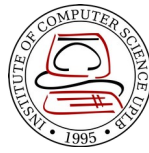
History



- *University of California at Los Angeles (UCLA), University of California at Santa Barbara (UCSB), Stanford Research Institute (SRI), and University of Utah* connected in December 1969 via software called **Network Control Protocol (NCP)**



History



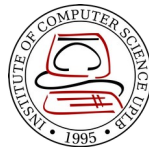
- 1972 – Vint Cerf and Bob Kahn worked on the [Internetting Project](#)
- 1973 – paper on [Transmission Control Protocol](#) (TCP)
 - Protocol to achieve end-to-end delivery, encapsulation, the datagram and functions of a gateway
- Original TCP was split into two: TCP and IP
 - TCP responsible for higher level functions such as segmentation, reassembly and error detection
 - IP handles datagram routing

History



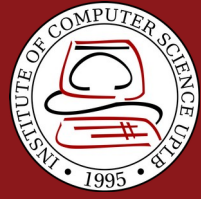
- 1981 – [Computer Science Network](#) (CSNET) was created for those ineligible to join ARPANET
- 1983 – ARPANET split into two networks: [Military Network](#) (MILNET) and ARPANET (nonmilitary use)
- 1986 – NSF sponsored the [National Science Foundation Network](#) (NSFNET), a backbone that connected 5 supercomputer centers located throughout the US
 - Officially replaced ARPANET in 1990
- 1991 – IBM, Merit and Verizon formed Advanced Network & Services (ANS) and build a new high speed Internet backbone called [Advanced Network Service Network](#) (ANSNET)

The Internet: Today



Set of peer networks that provides services to the world

- World Wide Web – invented at CERN by Tim Berners-Lee
- Multimedia – voice over IP (telephony), video over IP (like Skype), view sharing (like YouTube), television over IP (like PPLive)
- Peer-to-peer applications



Autonomous Systems

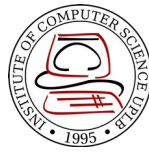
Autonomous System (AS)



- Collection of routers whose prefixes and routing policies are under common administrative control (SINGLE and CLEARLY DEFINED routing policy) [RFC 1930]
- Could be an ISP, a large company, a group of companies or a university
- Represents a connected group of one or more blocks of IP addresses, called **IP prefixes**, that have been assigned to that organization and provides a single routing policy to systems outside the AS

Ex. AS45209 – UPLB's ASN, with IPv4 prefix 202.92.144.0/22

Peering vs Transit

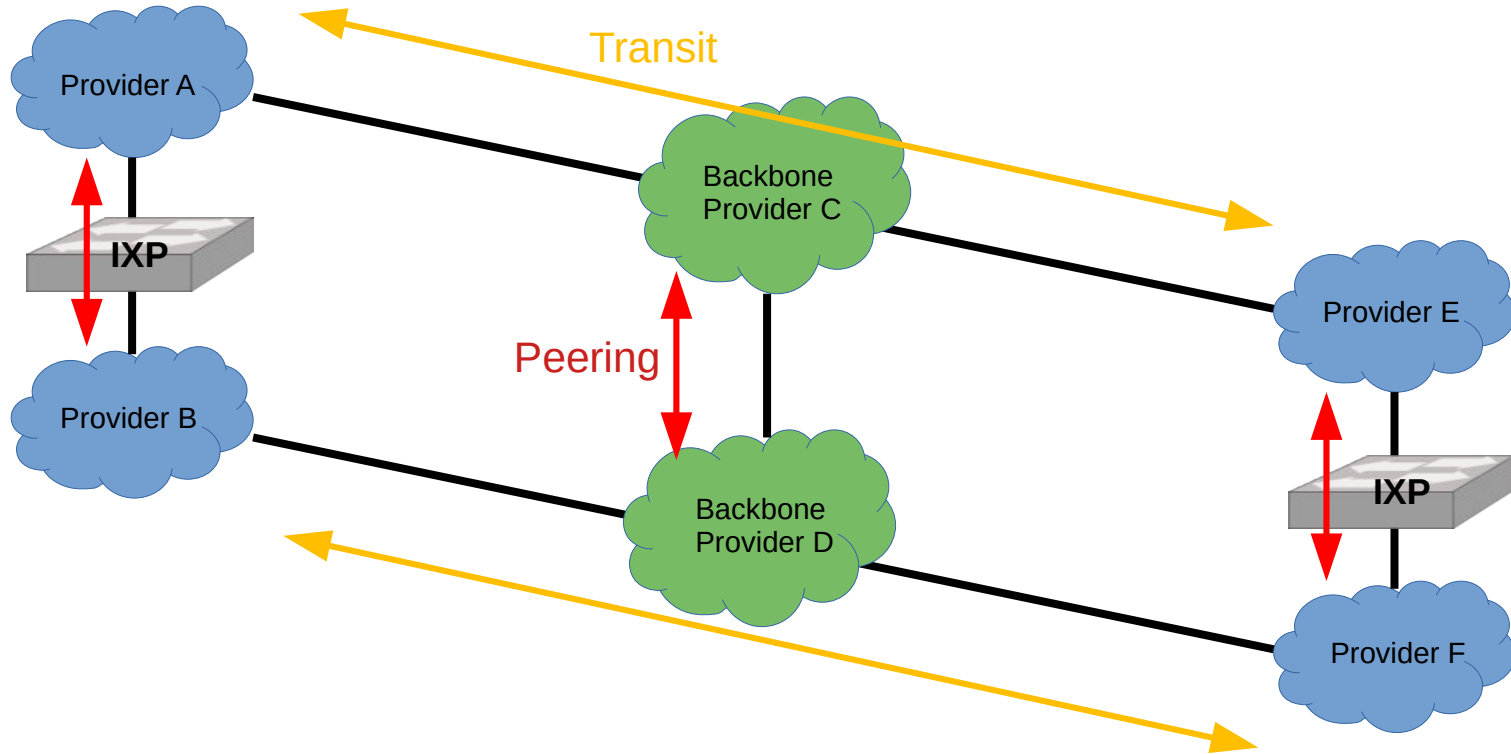


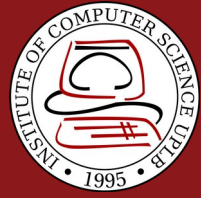
Peering – two or more autonomous networks interconnect directly with each other to exchange traffic

Transit – one autonomous network agrees to carry traffic that flows between another autonomous network and all other network

In peering, two ISPs only exchange the routes of their downstream customers and neither can see the upstream routes over peering connection. Because transit is a paid service, the provider will provide routes to the rest of the Internet, including their downstream partners, other peers and upstream providers

Peering vs Transit





Submarine Cables

Submarine Cables



First laid in the 1850s carrying telegraphy traffic, establishing the first instant links between continents. Subsequent generations carried telephone traffic then data communications traffic

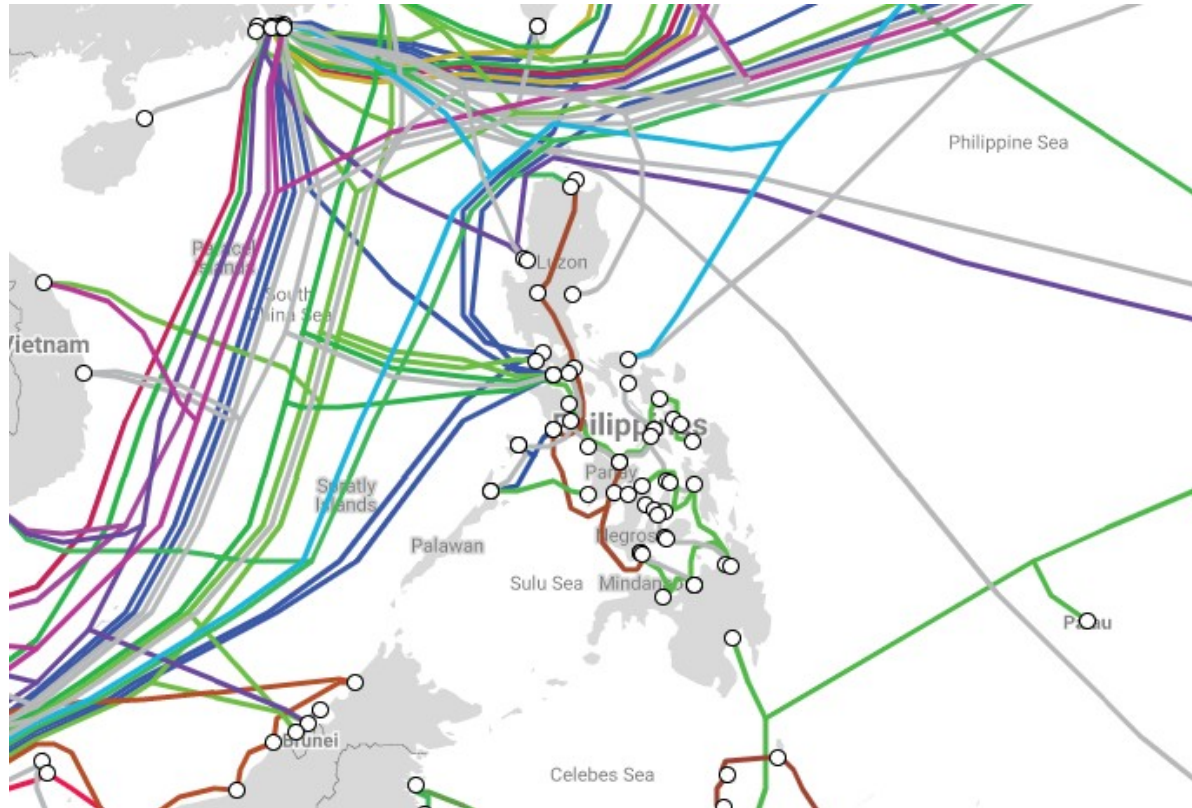


https://commons.wikimedia.org/wiki/File:Wolfe_Island_Wind_Project_Submarine_Power_Cable.jpg



https://commons.wikimedia.org/wiki/File:Diver_Checking_Underwater_Protection_of_Cable_-_Flickr_-_The_Official_CTBTOT_Photostream.jpg

Submarine Cable Map



<https://www.submarinecablemap.com/>

Asia-Pacific Cable Network 2



Submarine Cable List

APCN-2

[Email link](#)

RFS: 2001 December

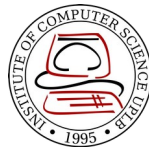
Cable Length: 19,000 km

Owners: Singtel, Verizon, KDDI, Chunghwa Telecom, AT&T, BT, Orange, Softbank Corp, NTT, Tata Communications, Telekom Malaysia, Starhub, PLDT, China Unicom, KT, Singtel Optus, Telstra, PCCW, China Telecom, LG Uplus, HKBN Enterprise Solutions, Vodafone
URL: n.a.

Landing Points

Batangas, Philippines
Busan, South Korea
Chikura, Japan
Chongming, China
Katong, Singapore
Kitaibaraki, Japan
Kuantan, Malaysia
Lantau Island, China
Shantou, China
Tanshui, Taiwan

Southeast Asia Japan Cable



Sponsored in part by Equinix

Feedback [t](#) [f](#) [gith](#)

[Q Search](#)

[Submarine Cable List](#)

Southeast Asia Japan Cable (SJC)

[✉ Email link](#)

RFS: 2013 June

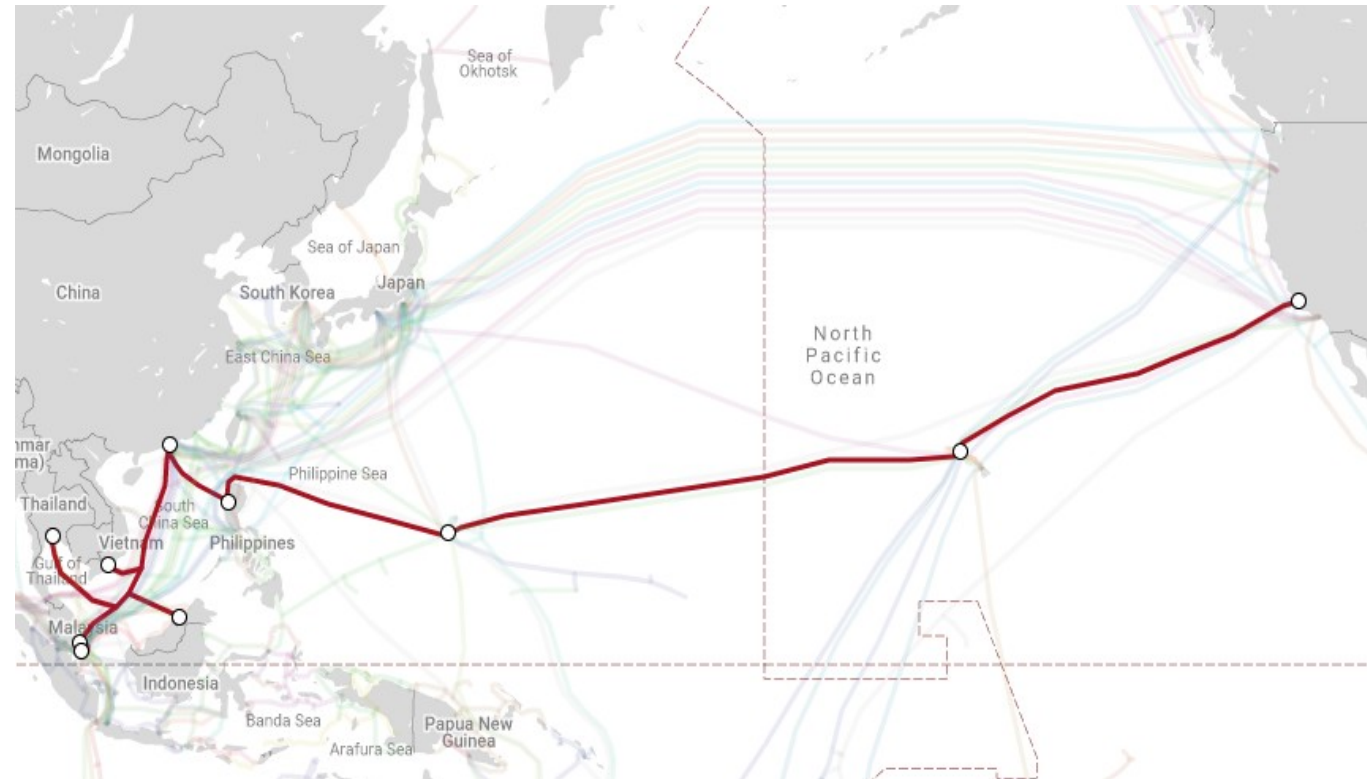
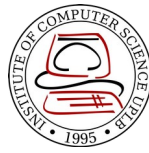
Cable Length: 8,900 km

Owners: Globe Telecom, Google, KDDI, Telkom Indonesia, Singtel, China Telecom, TOT, China Mobile, Chunghwa Telecom, Unified National Networks (UNN)
URL: n.a.

Landing Points

[Chikura, Japan](#)
[Chung Hom Kok, China](#)
[Nasugbu, Philippines](#)
[Shantou, China](#)
[Telisai, Brunei](#)
[Tuas, Singapore](#)

Asia-America Gateway



Asia-America Gateway (AAG) Cable System

[Email link](#)

RFS: 2009 November

Cable Length: 20,000 km

Owners: Telekom Malaysia, AT&T, Starhub, PLDT, CAT Telecom Public Company Limited, Airtel (Bharti), Telstra, Telkom Indonesia, BT, Eastern Telecom, Indosat Ooredoo, Spark New Zealand, Viettel Corporation, Saigon Postel Corporation, VNPT International, Brunei International Gateway, BayanTel, Ezecom

URL: <http://www.asia-america-gateway.com>

Landing Points

[Changi North, Singapore](#)

[Keawaula, HI, United States](#)

[La Union, Philippines](#)

[Lantau Island, China](#)

[Mersing, Malaysia](#)

[San Luis Obispo, CA, United States](#)

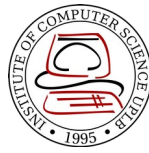
[Sriracha, Thailand](#)

[Tanguisson Point, GU, United States](#)

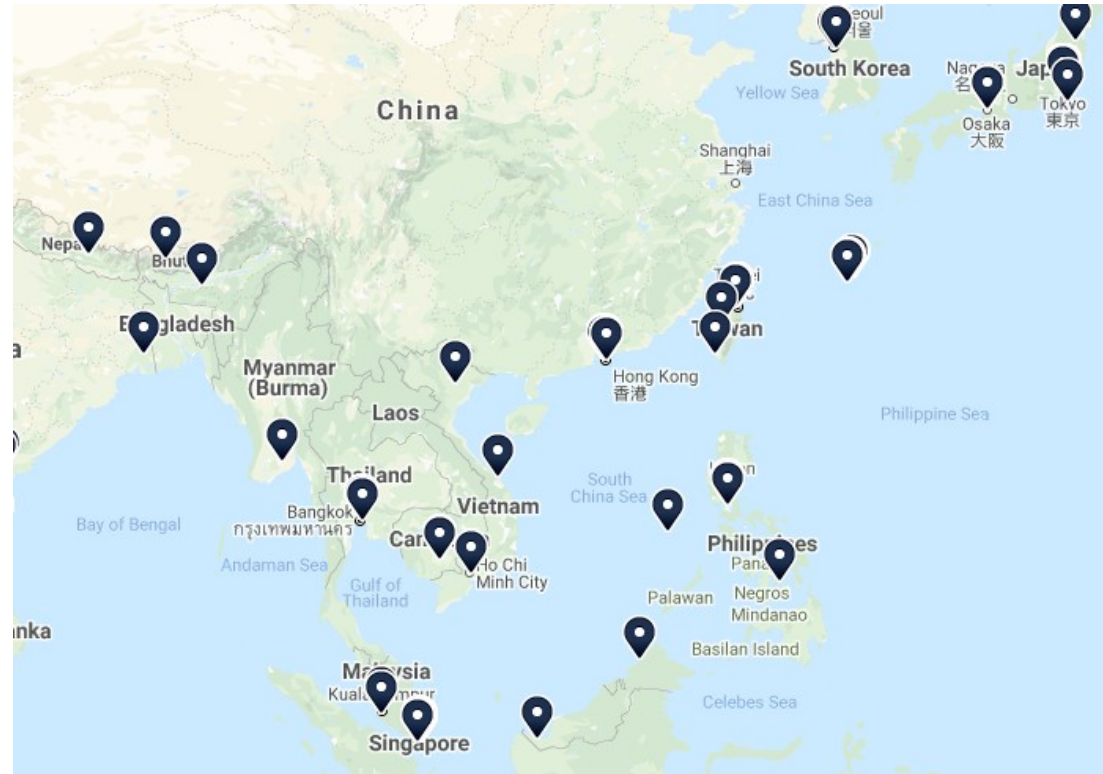
[Tungku, Brunei](#)

[Vung Tau, Vietnam](#)

Internet Exchange Points (IXP)

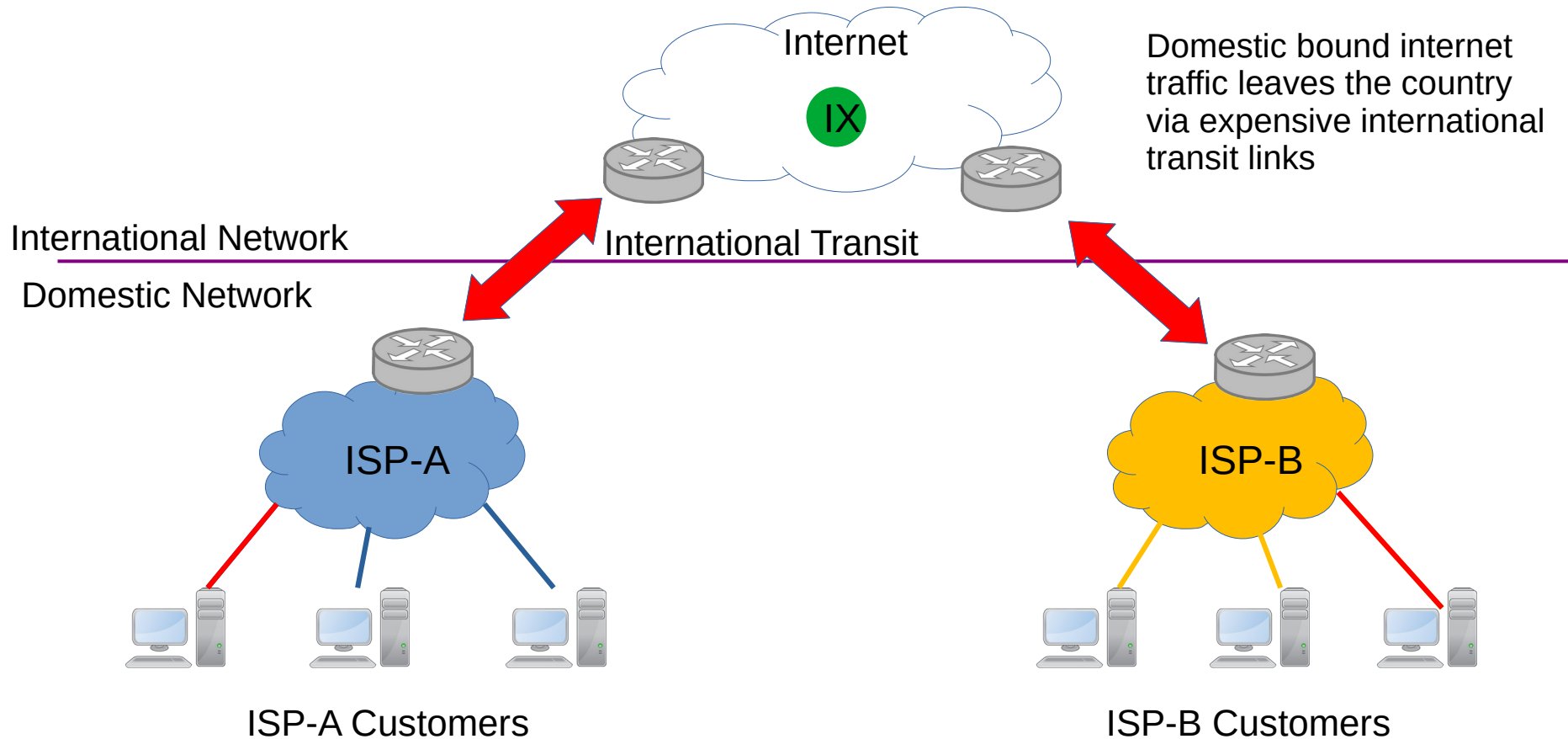


Physical location through which Internet infrastructure companies such as ISPs and content delivery networks (CDNs) connect with each other

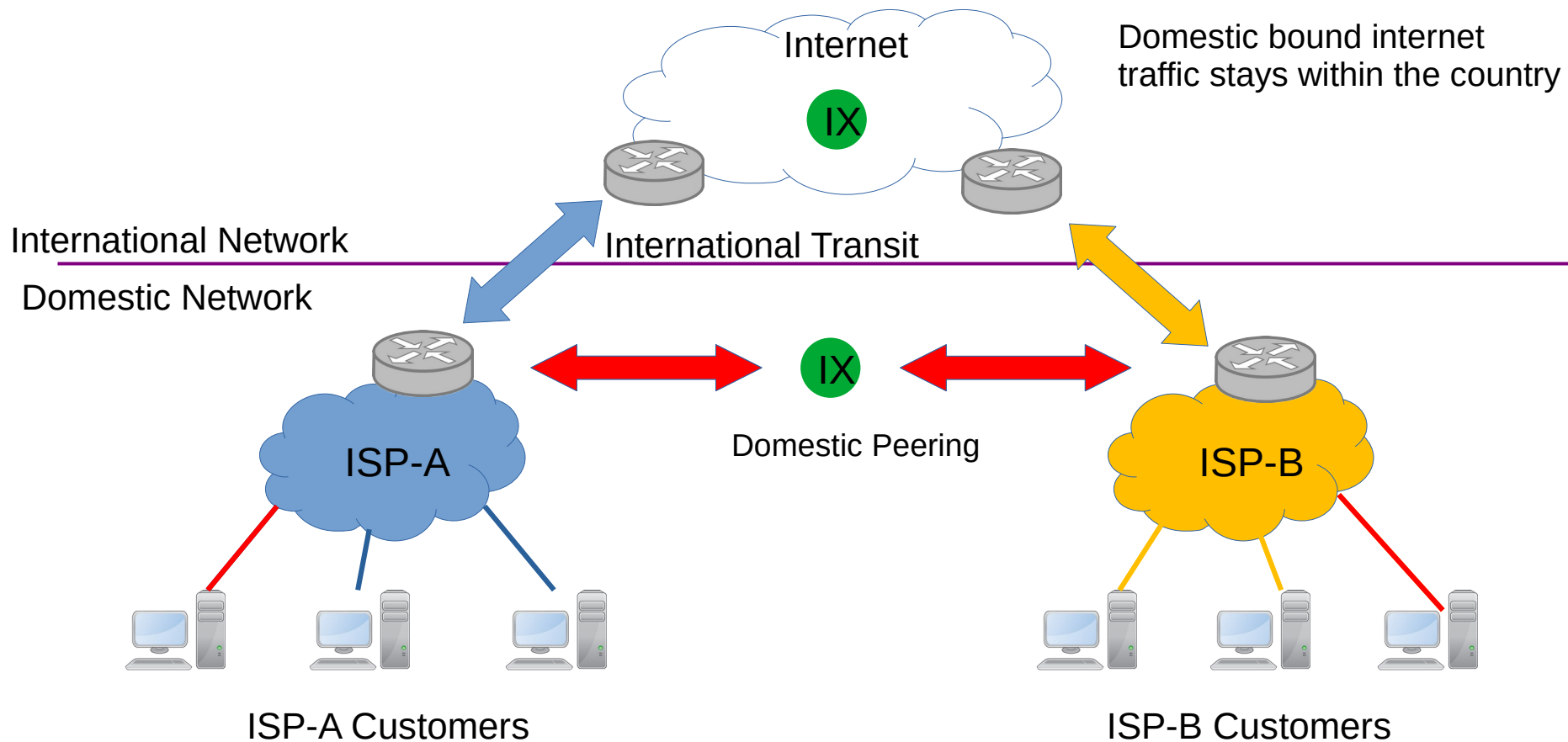


<https://www.internetexchangemap.com>

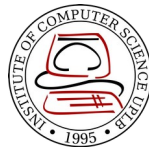
Without a Local IXP



With a Local IXP



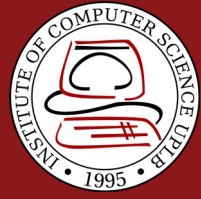
In the Philippines



Philippine Research, Education and Government Information Network (PREGINET) – national research and education network (NREN) that interconnects academic, research and government institutions
[<http://pregi.net/>]

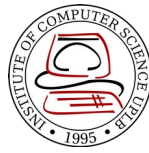
Philippine Open Internet Exchange (PHOpenIX) – only neutral Internet Exchange in the Philippines
[<http://phopenix.net>]

Both are being managed by DOST - ASTI



Standards

Protocol



Set of rules that govern data communications

- What is communicated, how it is communicated, when it is communicated

Elements

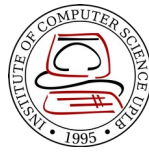
- **Syntax** – structure and format of a data
- **Semantics** – meaning of each section of bits
- **Timing** – when data should be sent and how fast they can be sent

Standards



- Creates and maintains an open and competitive market for manufacturers
- Guarantees national and international operability of data and telecommunications technology and processes
- **De facto** (“by fact”) – have not been approved by an organized body but have been adopted as standards through widespread use
- **De jure** (“by law”) – have been legislated by an officially recognized body

How are standards developed?



- Standard creation committees
 - ISO, ITU-T, CCITT, ANSI, IEEE, EIA
 - Slow approval process
- Forums – speeds up acceptance and use of technology
 - Conclusions presented in standards bodies
- Government regulatory agencies
 - FCC / NTC
 - To protect public interest

Internet Standard



- A thoroughly tested specification that is useful to and adhered to by those who work with the Internet
- Specification begins as an [Internet draft](#)
 - Work-in-progress, 6-month lifetime
- Upon recommendation, may be published as [Request for Comment](#)
 - Edited, assigned a number, made available to interested parties
 - Invented by [Steve Crocker](#) to provide record of Network Working Group's (NWG) design of the ARPANET
 - RFC 1 - “Host Software”
 - RFC 1000 - “Request for Comments Reference Guide”

RFC 1



Network Working Group
Request for Comments: 1

Steve Crocker
UCLA
7 April 1969

Crocker

RFC 1

Host Software

[Page 1]

7 April 1969

Title: Host Software
Author: Steve Crocker
Installation: UCLA
Date: 7 April 1969

Network Working Group Request for Comment: 1

CONTENTS

INTRODUCTION

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Messages

Links

IMP Transmission and Error Checking

Open Questions on the IMP Software

II. Some Requirements Upon the Host-to-Host Software

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Error Checking

III. The Host Software

Establishment of a Connection

High Volume Transmission

IV. Initial Experiments

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Experiment Two

Introduction

The software for the ARPA Network exists partly in the IMPs and partly in the respective HOSTs. BB&N has specified the software of the IMPs and it is the responsibility of the HOST groups to agree on HOST software.

During the summer of 1968, representatives from the initial four sites met several times to discuss the HOST software and initial experiments on the network. There emerged from these meetings a working group of three, Steve Carr from Utah, Jeff Rulifson from SRI, and Steve Crocker of UCLA, who met during the fall and winter. The most recent meeting was in the last week of March in Utah. Also present was Bill Duvall of SRI who has recently started working with Jeff Rulifson.

Somewhat independently, Gerard DeLoche of UCLA has been working on the HOST-IMP interface.

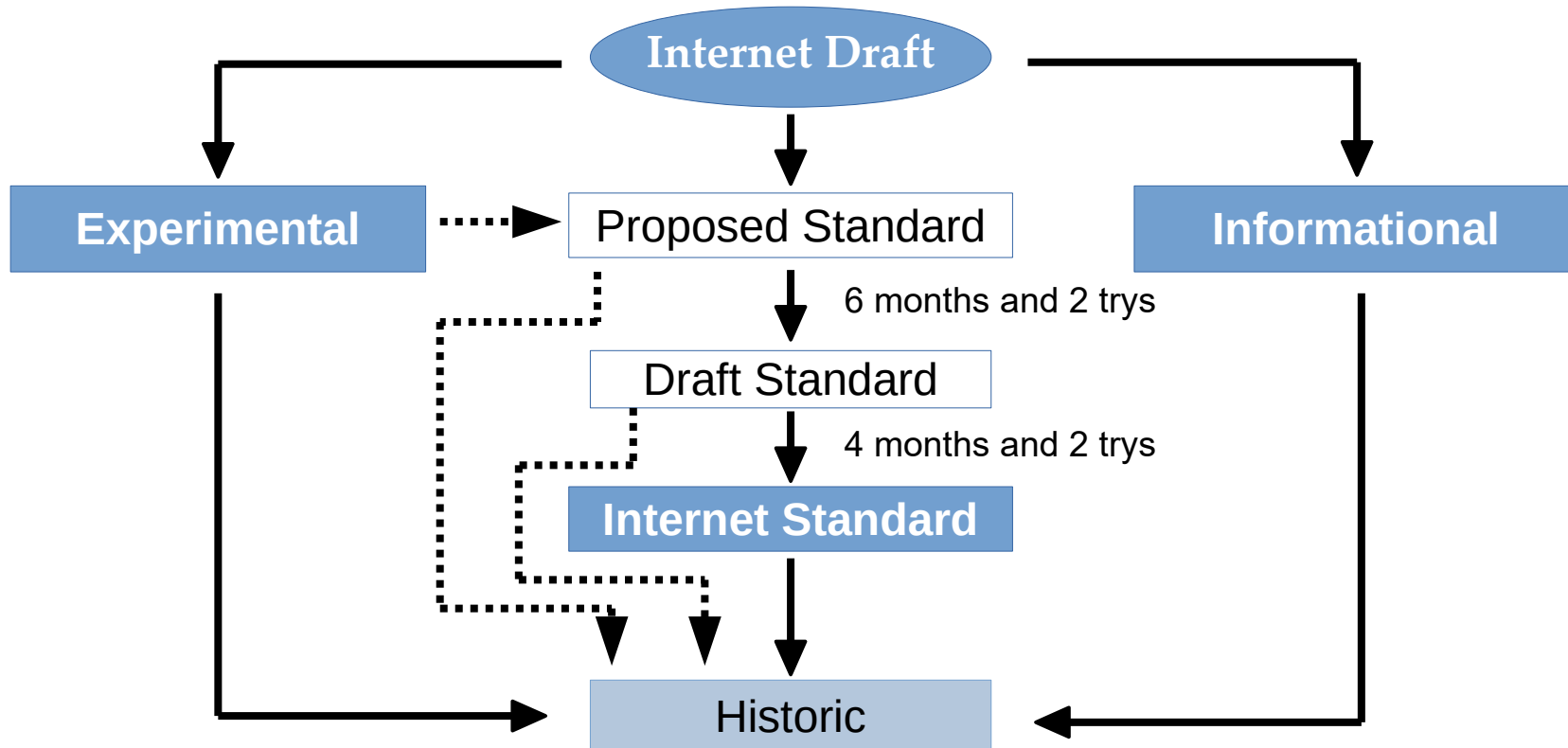
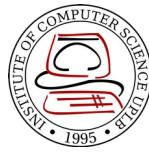
I present here some of the tentative agreements reached and some of the open questions encountered. Very little of what is here is firm and reactions are expected.

I. A Summary of the IMP Software

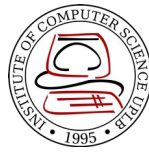
Messages

Information is transmitted from HOST to HOST in bundles called messages. A message is any stream of not more than 8080 bits, together with its header. The header is 16 bits and contains the following information:

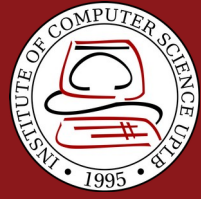
Maturity Levels of an RFC



RFC Requirements Level



- **Required** – it must be implemented by all Internet systems to achieve minimum performance (Example: IP, ICMP)
- **Recommended** – not required for minimum conformance; it is recommended for its usefulness (Example: FTP, TELNET)
- **Elective** – not required and not recommended, however a system can use it for its own benefit
- **Limited Use** – should only be used in limited situation; most experimental RFC fall under this
- **Not recommended** – inappropriate for general use; normally historic (deprecated) RFC fall under this



Administration

Internet Administration



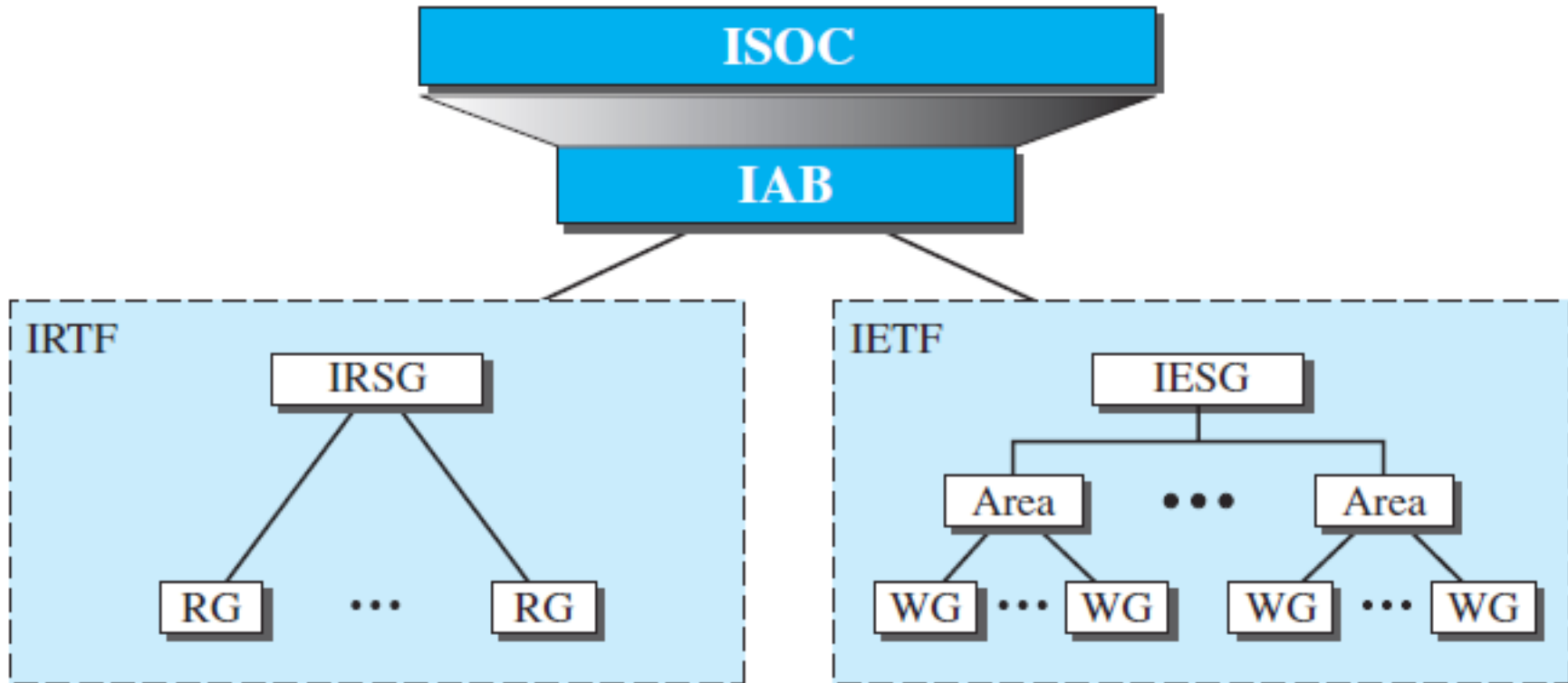
Internet Society (ISOC) – international, nonprofit organization formed in 1992 to provide support for the Internet standard process

Internet Architecture Board (IAB) – technical advisor of ISOC. Oversee the continuing development of the TCP/IP Protocol Suite

Internet Engineering Task Force (IETF) – forum of working groups managed by the Internet Engineering Steering Group (IESG). Responsible for identifying operational problems and proposing solutions to these problems

Internet Research Task Force (IRTF) – forum of working groups managed by the Internet Research Steering Group (IRSG). Focuses on long-term research topics

Internet Administration



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